

SHABLA, Bulgaria

WMO: 155610

Lat: 43.533N

Lon: 28.533E

Elev: 92

StdP: 14.65

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.0	20.0	10.5	9.5	16.7	14.6	11.6	21.1	36.7	25.4	36.1	24.1	20.8	320	0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.2	84.2	75.1	82.4	73.6	80.7	72.5	77.2	82.4	75.4	80.7	73.8	79.3	11.1	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.4	133.3	81.7	73.5	125.2	80.0	71.7	117.5	78.5	40.7	82.5	39.0	80.8	37.4	79.5	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.3	27.1	23.1	DB	11.0	89.0	6.0	3.8	6.6	91.8	3.1	94.0	-0.3	96.1	-4.7	98.9	(4)
(5)			WB	10.5	78.9	5.8	1.7	6.3	80.1	2.9	81.1	-0.4	82.0	-4.6	83.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.0	35.9	37.4	42.5	49.2	58.7	67.3	72.5	72.9	65.6	57.2	48.0	39.2
	DBStd	14.51	8.46	7.98	6.40	5.00	4.88	4.26	3.78	3.79	4.91	6.62	7.55	8.73
	HDD50	1600	439	356	244	70	3	0	0	0	0	19	124	345
	HDD65	4684	902	773	696	473	199	26	1	53	250	511	799	
	CDD50	3044	1	2	12	48	273	518	697	710	467	242	63	11
	CDD65	655	0	0	0	0	4	94	233	246	70	8	0	0
	CDH74	4005	0	0	1	0	7	385	1644	1802	158	8	0	0
	CDH80	519	0	0	0	0	0	21	229	265	4	0	0	0
Wind	WSAvg	12.0	13.1	13.6	13.1	11.1	10.2	10.2	10.5	10.9	12.0	12.9	12.7	13.9
Precipitation	PrecAvg	17.9	1.2	0.9	1.2	1.3	1.4	1.4	1.5	1.3	2.7	1.9	1.5	1.7
	PrecMax	27.8	3.2	2.4	3.7	4.8	3.2	3.9	7.0	4.1	11.6	9.2	3.7	5.7
	PrecMin	9.4	0.1	0.1	0.1	0.1	0.2	0.0	0.1	0.0	0.2	0.0	0.2	0.2
	PrecStd	4.9	0.8	0.6	0.8	1.0	0.8	1.0	1.6	1.2	2.6	1.8	0.9	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.7	58.4	64.8	68.3	74.0	82.6	86.5	87.1	79.6	74.6	66.1	58.9
		MCWB	50.0	49.6	53.3	58.1	65.1	73.2	73.8	76.5	68.4	67.0	59.6	54.5
	2%	DB	51.6	52.3	57.6	62.7	70.7	79.7	83.9	84.9	77.4	71.2	63.2	56.2
		MCWB	48.0	46.7	49.3	54.9	64.0	71.6	75.0	75.9	69.0	65.5	59.1	52.8
	5%	DB	49.1	49.2	53.4	58.8	68.9	77.9	82.4	82.8	75.8	69.2	60.6	53.8
		MCWB	46.0	45.3	47.8	53.6	63.3	70.6	73.8	74.1	68.2	64.3	57.1	51.0
	10%	DB	47.0	47.1	50.8	56.6	67.3	76.2	80.9	81.0	74.3	67.2	58.4	51.6
		MCWB	44.5	43.9	46.6	52.4	62.1	69.5	73.0	72.7	67.4	62.5	55.0	48.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.8	50.3	53.8	59.2	67.1	75.4	78.7	80.0	72.6	69.0	61.6	55.3
		MCDB	55.2	56.6	64.1	65.2	71.1	80.1	83.4	85.3	77.2	72.8	64.3	57.7
	2%	WB	48.3	47.5	50.2	56.1	65.4	73.4	77.2	77.8	70.9	66.7	59.2	53.2
		MCDB	51.0	51.5	55.7	61.3	69.4	77.9	82.2	83.5	75.5	70.2	62.4	55.8
	5%	WB	46.4	45.8	48.5	54.2	63.9	71.8	75.6	75.7	69.7	64.8	57.3	51.0
		MCDB	48.7	48.7	52.7	58.1	68.1	76.8	80.9	81.1	74.4	68.4	60.2	53.5
	10%	WB	44.5	44.2	47.0	52.7	62.5	70.2	74.0	74.0	68.4	62.9	55.5	48.9
		MCDB	46.8	46.9	50.5	56.1	66.7	75.2	79.7	79.6	73.2	66.5	58.2	51.3
Mean Daily Temperature Range	5% DB	MDBR	10.2	10.3	10.5	10.5	12.0	13.2	13.9	14.2	12.6	11.1	10.6	10.7
		MCDBR	12.4	13.5	15.1	14.7	14.0	15.3	15.5	15.8	14.3	12.9	11.4	11.1
		MCWBR	9.6	9.5	9.4	9.8	10.0	10.7	10.8	11.3	9.8	9.7	8.6	9.0
	5% WB	MCDBR	11.3	12.5	12.9	12.3	12.5	13.6	13.3	13.6	11.8	11.3	10.6	9.9
		MCWBR	9.4	9.2	8.7	8.9	9.8	10.9	10.9	11.0	9.5	9.4	8.6	8.2
Clear-Sky Solar Irradiance	taub	0.358	0.387	0.417	0.458	0.430	0.432	0.441	0.451	0.431	0.400	0.377	0.360	
	taud	2.324	2.243	2.187	2.088	2.258	2.292	2.278	2.242	2.266	2.321	2.322	2.322	
	Ebn at Noon	238	250	258	256	268	267	263	256	251	243	228	224	
	Edh at Noon	27	34	41	49	42	41	41	42	37	31	27	25	
All-Sky Solar Radiation	RadAvg	415	656	1069	1546	1966	2151	2141	1919	1367	852	499	374	
	RadStd	64	80	112	150	108	132	117	92	113	78	63	61	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)	+0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-205	+189	+110

Nomenclature: See separate page